

REXMI - Physical AI for space

The first robot

A wheeled quadruped designed for lunar surface work, with crater access built into the system concept. Wheels cover traversable ground; legs adapt clearance and contact placement across broken terrain. Reinforcement learning and onboard navigation connect movement to mission execution.

Reach the site. Collect the evidence. Return useful data.

Mission roles

Scout and recon

Provide navigational data: terrain maps, hazards, traversable routes and points of interest. Combine route information with available resource maps to help mission teams plan where to go and how to get there.

Example: Survey the route from a lander to a candidate resource site, flag difficult slopes and identify an alternative approach.

Resource mapping & crater access

Reach promising ISRU sites identified from orbit, including selected crater slopes and interiors. Planned spectrometer and ground-penetrating radar payloads will add local composition and subsurface measurements to resource maps. Polar crater ice is a key target. [1-3]

Example: Traverse a candidate ice-bearing area, collect location-tagged measurements and help validate orbital interpretations before excavation equipment, power and mission time are committed.

Supplies

Carry mission-sized instruments, equipment and tools to reduce astronaut workload. Payload configurations will support transport between landing areas, work sites and surface infrastructure.

Example: Bring a science instrument and tool kit to an astronaut work site, freeing the crew to focus on the task.

Inspection

Revisit infrastructure and scientific instruments to document condition and changes. Repeatable positioning and onboard sensing will support comparisons across visits.

Example: Follow a repeat survey route around a lander, recording visible dust buildup, damage or obstructions for operator review.

Current foundation

REXMI's development includes reinforcement-learning mobility, navigation, movement-skill switching and recovery in simulation. Crater access is a current design objective. Physical hardware validation and lunar environmental qualification are planned development work. [4]

Development plans

Five connected workstreams support the mission roles.

Lunar mobility simulation

Planned training and evaluation cover reduced gravity, loose terrain, traction variation and actuator limits. Crater scenarios include approach, descent, traverse and exit, with stability, route completion, energy use and recovery as evaluation measures.

Adaptive thermal control

The thermal architecture will combine heat paths, radiators, insulation, heaters and temperature-aware operation. Planned analysis links payload and actuator heat to terrain, illumination and mission duration, including the demands of shadowed crater operations.

Perception

Planned onboard sensing and localization will support terrain mapping, hazard detection, traversability assessment and precise positioning at science targets. Instrument observations will be associated with location, time and acquisition conditions for scientific interpretation.

Dust protection

Mechanism development will address shields, cable routing, bearing seals and lubrication, alongside protection for optics and radiators. Planned testing covers dust exposure, wear, vacuum and thermal cycling.

Robot validation

Planned hardware trials will evaluate mobility, instrument operation, load carrying and inspection on representative terrain. Validation will measure repeatability, data quality, energy use and operator intervention, with protective responses to lost localization, communication faults and thermal limits.

Mission and payload integration

The system plan includes payload mounting, power and data interfaces, instrument calibration, communications and onboard data storage. Mission design will define mass, energy reserves, thermal limits and return paths. Access to permanent shadow requires its own power, thermal and communications provisions.

Long-term vision: physical intelligence for space, across quadrupeds, humanoids and new hybrids.

Sources

1. NASA: [Moon Water and Ices](#). Polar ice and volatiles.
2. NASA: [Lunar Terrain Vehicle instruments](#), 10 July 2025. Spectroscopy and GPR science roles.
3. NASA: [Lunar surface science payloads](#). Local measurements to ground-truth orbital observations.
4. REXMI project descriptions, August 2026; founder design direction, September 2026. Development status and plans.